Epic-FHIR — API category: Clinical Imaging (sections 11 – 11.1.4)

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11 API category: Clinical Imaging

File: site/apis/11-Clinical-Imaging/index.html

R4 (4)

API | Method | URL template | Type
Observation.Create (DICOM Image Characteristics) (R4) | POST | /api/FHIR/R4/Observation | Industry-Standard
Observation.Read (DICOM Image Characteristics) (R4) | GET | /api/FHIR/R4/Observation/{ID} | Industry-Standard
Observation.Search (DICOM Image Characteristics) (R4) | GET | /api/FHIR/R4/Observation?based-on={based-on}&category={category}&code={code}&date={date}&focus={focus}&issued={issued}&patient={patient}&subject={subject}&_count={_count}&_id={_id}&_include={_include}&_revinclude={_revinclude} | Industry-Standard
Observation.Update (DICOM Image Characteristics) (R4) | PUT | /api/FHIR/R4/Observation/{ID} | Industry-Standard

11.1 Clinical Imaging: R4

File: site/apis/11-Clinical-Imaging/1-R4/index.html

* Observation.Create (DICOM Image Characteristics) (R4)
* Observation.Read (DICOM Image Characteristics) (R4)
* Observation.Search (DICOM Image Characteristics) (R4)
* Observation.Update (DICOM Image Characteristics) (R4)

11.1.1 Observation.Create (DICOM Image Characteristics) (R4)

File: site/apis/11-Clinical-Imaging/1-R4/11224-Observation-Create-DICOM-Image-Characteristics-R4.html
Epic API ID | 11224
FHIR version | R4
Categories | Clinical Imaging
API type | Industry-Standard
API group | DICOM Image Characteristics Group
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | POST
URL template | /api/FHIR/R4/Observation
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11224

Description

The FHIR Observation resource defines measurements and assertions about a patient. This resource allows clients to file imaging metadata into the database, specifically CT Dose and Image Quality Category through codeable concepts, Calculated CT size-adjusted dose through component codes, and Calculated CT global noise through component codes.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False | | | An instance of the R4 Observation resource.
Observation.category | CodeableConcept | False | Yes | | “imaging” to create DICOM Image Characteristics observation.
Observation.code | CodeableConcept | False | | | LOINC code. For Excessive Radiation (ExRad) integration, this is always LOINC#96914-7 “CT dose and image quality category”.
Observation.component | BackboneElement | True | Yes | | Components as they correspond to Global Noise or Size-Adjusted Dose.
Observation.component.code | CodeableConcept | False | | | LOINC code. For the Excessive Radiation (ExRad) integration, this will be LOINC#96912-1 for Calculated CT global noise or LOINC#96913-9 for Calculated CT size-adjusted dose.
Observation.component.valueQuantity | Quantity | False | | | The finding for each component code.
Observation.effectiveDateTime | dateTime | False | | | Clinically relevant instant for observation.
Observation.focus | Reference | False | Yes | | Reference to a DiagnosticReport resource. Use the DiagnosicRepot.Search (Results) API to obtain a DiagnosticReport for the target patient.
Observation.status | code | False | | | The status of the observation. Only a value of “final” is supported. We do not support filing data that isn’t finalized.
Observation.subject | Reference | False | | | Reference to a patient resource for the patient associated with this observation.
Observation.valueCodeableConcept | CodeableConcept | True | | | Value of the observation, such as the actual result within LOINC#96914-7 as selected from the specified LOINC answer (LA) list.

Sample request

```
{
  "resourceType": "Observation",
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "imaging",
          "display": "Imaging"
        }
      ]
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "96914-7",
        "display": "CT dose and image quality category"
      }
    ],
    "text": "CT dose and image quality category"
  },
  "subject": {
    "reference": "Patient/e1n5j3r6Goo1BGu7ckMTnSw3"
  },
  "focus": [
    {
      "reference": "DiagnosticReport/e9BvJll1zDW79iDWWfs2fTwXfc8vLgyjQb7F3I38cgZ43"
    }
  ],
  "encounter": {
    "reference": "Encounter/eknUIpa6enj2w4Kbl5CYa93rRgqcM7EeV5j.3hhPfshg3"
  },
  "effectiveDateTime": "2025-01-27T20:38:30Z",
  "valueCodeableConcept": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "LA31753-9",
        "display": "Abdomen and Pelvis Routine Dose"
      }
    ]
  },
  "component": [
    {
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "96912-1",
            "display": "Calculated CT global noise"
          }
        ]
      },
      "valueQuantity": {
        "value": 20.01,
        "unit": "[hnsf'U]",
        "system": "http://unitsofmeasure.org",
        "code": "[hnsf'U]"
      }
    },
    {
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "96913-9",
            "display": "Calculated CT size-adjusted dose"
          }
        ]
      },
      "valueQuantity": {
        "value": 237.54,
        "unit": "mGy.cm",
        "system": "http://unitsofmeasure.org",
        "code": "mGy.cm"
      }
    }
  ]
}
```


Sample response

```
HTTP 201
```

11.1.2 Observation.Read (DICOM Image Characteristics) (R4)

File: site/apis/11-Clinical-Imaging/1-R4/11227-Observation-Read-DICOM-Image-Characteristics-R4.html
Epic API ID | 11227
FHIR version | R4
Categories | Clinical Imaging
API type | Industry-Standard
API group | DICOM Image Characteristics Group
USCDI | No
Keywords | DICOM
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11227

Description

The FHIR Observation resource defines measurements and assertions about a patient. This resource allows clients to file imaging metadata into the database, specifically CT Dose and Image Quality Category through codeable concepts, Calculated CT size-adjusted dose through component codes, and Calculated CT global noise through component codes.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | False | | | The Observation FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False | | | An instance of the R4 Observation resource.
Observation.category | CodeableConcept | False | Yes | | Always "imaging" for the DICOM Image Characteristics API.
Observation.code | CodeableConcept | False | | | LOINC code. For Excessive Radiation (ExRad) integration, this will always be LOINC#96914-7 “CT dose and image quality category”.
Observation.component | BackboneElement | True | Yes | | Components as they correspond to Global Noise or Size-Adjusted Dose.
Observation.component.code | CodeableConcept | False | | | LOINC code. For Excessive Radiation (ExRad) integration, this will be LOINC#96912-1 for Calculated CT global noise or LOINC#96913-9 for Calculated CT size-adjusted dose.
Observation.component.valueQuantity | Quantity | False | | | The finding for this component code.
Observation.effectiveDateTime | dateTime | False | | | Clinically relevant instant for observation.
Observation.focus | Reference | False | Yes | | Reference to a DiagnosticReport resource.
Observation.id | id | False | | | The FHIR ID.
Observation.status | code | False | | | The status of the observation. Only a value of “final” is supported.
Observation.subject | Reference | False | | | Reference to a patient resource.
Observation.valueCodeableConcept | CodeableConcept | True | | | Value of the observation, such as the actual result within LOINC#96914-7 as selected from the specified LOINC answer (LA) list.

Sample request

```
/api/FHIR/R4/Observation/fyTXzRRro10FweREciN-APA4
```

Sample response


```
{
  "resourceType": "Observation",
  "id": "fyTXzRRro10FweREciN-APA4",
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "imaging",
          "display": "Imaging"
        }
      ]
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "96914-7",
        "display": "CT dose and image quality category"
      }
    ],
    "text": "CT dose and image quality category"
  },
  "subject": {
    "reference": "Patient/eshFiUIi8bw98dvN8-8PwFA3",
```

```
    "display": "Foley, FHIROne"
  },
  "focus": [
    {
      "reference": "DiagnosticReport/e6qlhmNceq5U-OVhITNEAnZ8l1.k5BJxs5pcIsWSbAOQ3"
    }
  ],
  "encounter": {
    "reference": "Encounter/eWjDXw8vc57bDoXTNfql.H8g163udXkLtkfUyd4B4QsY3",
    "identifier": {
      "use": "usual",
      "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.3.698084.8",
      "value": "10004900788"
    },
    "display": "RIS Ancillary Orders"
  },
  "effectiveDateTime": "2025-01-27T20:38:30Z",
  "valueCodeableConcept": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "LA31753-9",
        "display": "Abdomen and Pelvis Routine Dose"
      }
    ]
  },
  "component": [
    {
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "96912-1",
            "display": "Calculated CT global noise"
          }
        ]
      },
      "valueQuantity": {
        "value": 20.01,
        "unit": "[hnsf'U]",
        "system": "http://unitsofmeasure.org",
        "code": "[hnsf'U]"
      }
    },
    {
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "96913-9",
            "display": "Calculated CT size-adjusted dose"
          }
        ]
      },
      "valueQuantity": {
        "value": 237.54,
        "unit": "mGy.cm",
        "system": "http://unitsofmeasure.org",
        "code": "mGy.cm"
      }
    }
  ]
}
```


11.1.3 Observation.Search (DICOM Image Characteristics) (R4)

File: site/apis/11-Clinical-Imaging/1-R4/11226-Observation-Search-DICOM-Image-Characteristics-R4.html
Epic API ID | 11226
FHIR version | R4
Categories | Clinical Imaging
API type | Industry-Standard
API group | DICOM Image Characteristics Group
USCDI | No
Keywords | DICOM
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Observation?based-on={based-on}&category={category}&code={code}&date={date}&focus={focus}&issued={issued}&patient={patient}&subject={subject}&_count={_count}&_id={_id}&_include={_include}&_revinclude={_revinclude}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11226

Description

The FHIR Observation resource defines measurements and assertions about a patient. This resource allows clients to file imaging metadata into the database, specifically CT Dose and Image Quality Category through codeable concepts, Calculated CT size-adjusted dose through component codes, and Calculated CT global noise through component codes.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
based-on | String | True | Yes | | Unsupported
category | token | Either code or category is required. | Yes | | Use a category of "imaging" for this API.
code | token | Either code or category is required. | Yes | | Use a code of 96914-7 to search for results of this type.
combo-code | token | True | Yes | | LOINC code that is present in the code or component.code response element.
combo-data-absent-reason | token | True | Yes | | Unsupported
combo-value-concept | token | True | Yes | | The value of the observation if valueCodeableConcept or component.valueCodeableConcept is populated.
component-code | token | True | Yes | | LOINC code that is present in the code or component.code response element.
component-data-absent-reason | token | True | Yes | | Unsupported
component-value-concept | token | True | Yes | | The value of the observation if valueCodeableConcept or component.valueCodeableConcept is populated.
data-absent-reason | token | True | Yes | | Unsupported
date | date | True | Yes | | The date range of the effective date of the imaging record.
derived-from | String | True | Yes | | Unsupported
device | String | True | Yes | | Unsupported
encounter | reference | True | Yes | | Reference to a patient encounter resource the observation is about.
focus | reference | True | Yes | | Reference to a DiganosticReport resource the observation is about.
has-member | reference | True | Yes | | Unsupported
identifier | token | True | Yes | | Unsupported
issued | date | True | Yes | | The date and time the observation was documented.
method | token | True | Yes | | Unsupported
part-of | String | True | Yes | | Unsupported
patient | reference | Either patient or subject is required. | Yes | | The patient associated with this observation.
performer | reference | True | Yes | | The practitioner who performed the diagnostic report.
specimen | String | True | Yes | | Unsupported
status | token | True | Yes | | The status of the observation (always "final").
subject | reference | Either patient or subject is required. | Yes | | The patient associated with this observation.
value-concept | token | True | Yes | | The value of the observation if valueCodeableConcept is populated.
value-date | date | True | Yes | | Unsupported
value-string | string | True | Yes | | Matches to valueString, if present.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.Observation | Observation | False | | | An instance of the R4 Observation resource.
Bundle.entry.resource.Observation.category | CodeableConcept | False | Yes | | Always "imaging" for the DICOM Image Characteristics API.
Bundle.entry.resource.Observation.code | CodeableConcept | False | | | LOINC code. For Excessive Radiation (ExRad) integration, this will always be LOINC#96914-7 “CT dose and image quality category”.
Bundle.entry.resource.Observation.component | BackboneElement | True | Yes | | Components as they correspond to Global Noise or Size-Adjusted Dose.
Bundle.entry.resource.Observation.component.code | CodeableConcept | False | | | LOINC code. For Excessive Radiation (ExRad) integration, this will be LOINC#96912-1 for Calculated CT global noise or LOINC#96913-9 for Calculated CT size-adjusted dose.
Bundle.entry.resource.Observation.component.valueQuantity | Quantity | False | | | The finding for this component code.
Bundle.entry.resource.Observation.effectiveDateTime | dateTime | False | | | Clinically relevant instant for observation.
Bundle.entry.resource.Observation.focus | Reference | False | Yes | | Reference to a DiagnosticReport resource.
Bundle.entry.resource.Observation.id | id | False | | | The FHIR ID.
Bundle.entry.resource.Observation.status | code | False | | | The status of the observation. Only a value of “final” is supported.
Bundle.entry.resource.Observation.subject | Reference | False | | | Reference to a patient resource.
Bundle.entry.resource.Observation.valueCodeableConcept | CodeableConcept | True | | | Value of the observation, such as the actual result within LOINC#96914-7 as selected from the specified LOINC answer (LA) list.

Sample request

```
/api/FHIR/R4/Observation?patient=eHDVcN3QnXpByseOB0H7qnw3&category=imaging&code=96914-7
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url":
    "https://hostname/instance/api/FHIR/R4/Observation?category=imaging&code=96914-7&patient=eHDVcN3QnXpByseOB0H7qnw3"
```

```
  }
],
"entry": [
  {
    "link": [
      {
        "relation": "self",
        "url": "https://hostname/instance/api/FHIR/R4/Observation/fwaHTvNN6WXBBCqRzpLzGbQ4"
      }
    ],
    "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/fwaHTvNN6WXBBCqRzpLzGbQ4",
    "resource": {
      "resourceType": "Observation",
      "id": "fwaHTvNN6WXBBCqRzpLzGbQ4",
      "status": "final",
      "category": [
        {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/observation-category",
              "code": "imaging",
              "display": "Imaging"
            }
          ]
        }
      ],
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "96914-7",
            "display": "CT dose and image quality category"
          }
        ],
        "text": "CT dose and image quality category"
      },
      "subject": {
        "reference": "Patient/eHDVcN3QnXpByseOB0H7qnw3",
        "display": "Foley, FHIRSensitive"
      },
      "focus": [
        {
          "reference": "DiagnosticReport/eaaUoMjuXumBEVTP6vXPNB1rnYbvPw-spaXR.EWm9Eh43"
        }
      ],
      "encounter": {
        "reference": "Encounter/erklGa4GUvFzdCaUMCUgYMZMxT7wXJGkUg4S2BYSf8GU3",
        "identifier": {
          "use": "usual",
          "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.3.698084.8",
          "value": "10004901097"
        },
        "display": "Office Visit"
      },
      "effectiveDateTime": "2025-03-03T19:13:32Z",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "LA31765-3",
            "display": "Extremity"
          }
        ]
      },
      "component": [
        {
          "code": {
            "coding": [
              {
                "system": "http://loinc.org",
                "code": "96912-1",
                "display": "Calculated CT global noise"
              }
            ]
          },
          "valueQuantity": {
            "value": 120,
            "unit": "[hnsf'U]",
            "system": "http://unitsofmeasure.org",
            "code": "[hnsf'U]"
          }
        },
        {
          "code": {
            "coding": [
              {
                "system": "http://loinc.org",
                "code": "96913-9",
                "display": "Calculated CT size-adjusted dose"
```

```
              }
            ]
          },
          "valueQuantity": {
            "value": 3500,
            "unit": "mGy.cm",
            "system": "http://unitsofmeasure.org",
            "code": "mGy.cm"
          }
        }
      ]
    },
    "search": {
      "mode": "match"
    }
  }
 ]
}
```

11.1.4 Observation.Update (DICOM Image Characteristics) (R4)

File: site/apis/11-Clinical-Imaging/1-R4/11225-Observation-Update-DICOM-Image-Characteristics-R4.html
Epic API ID | 11225
FHIR version | R4
Categories | Clinical Imaging
API type | Industry-Standard
API group | DICOM Image Characteristics Group
USCDI | No
Keywords | DICOM
Available in inpatient and other contexts | Yes
HTTP method | PUT
URL template | /api/FHIR/R4/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11225

Description

The FHIR Observation resource defines measurements and assertions about a patient. This resource allows clients to update imaging metadata into the database, specifically CT Dose and Image Quality Category through codeable concepts, Calculated CT size-adjusted dose through component codes, and Calculated CT global noise through component codes.

This resource uses an E-Tag and an If-Match HTTP header.

If the update is successful, the HTTP status code 200 is returned. If the update is not successful, an operation outcome is returned.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The FHIR ID of the of the resource you would like to update.
Observation | Observation | False | | | An instance of the R4 Observation resource.
Observation.category | CodeableConcept | False | Yes | | Use "imaging” to update DICOM Image Characteristics observations.
Observation.code | CodeableConcept | False | | | LOINC code. For Excessive Radiation (ExRad) integration, this is always LOINC#96914-7 “CT dose and image quality category”.
Observation.component | BackboneElement | True | Yes | | Components as they correspond to Global Noise or Size-Adjusted Dose.
Observation.component.code | CodeableConcept | False | | | LOINC code. For the Excessive Radiation (ExRad) integration, this is LOINC#96912-1 for Calculated CT global noise or LOINC#96913-9 for Calculated CT size-adjusted dose.
Observation.component.valueQuantity | Quantity | False | | | The finding for each component code.
Observation.effectiveDateTime | dateTime | False | | | Clinically relevant instant for observation.
Observation.focus | Reference | False | Yes | | Reference to a DiagnosticReport resource.
Observation.id | id | False | | | The FHIR ID of the of the resource you would like to update.
Observation.status | code | False | | | The status of the observation. Only a value of “final” is supported. We do not support filing data that isn’t finalized.
Observation.subject | Reference | False | | | Reference to the patient associated with this observation.
Observation.valueCodeableConcept | CodeableConcept | True | | | Value of the observation, such as the actual result within LOINC#96914-7 as selected from the specified LOINC answer (LA) list.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False | | | An instance of the R4 Observation resource.

Sample request

```
{
  "resourceType": "Observation",
```

```
  "id": "fyTXzRRro10FweREciN-APA4",
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "imaging",
          "display": "Imaging"
        }
      ]
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "96914-7",
        "display": "CT dose and image quality category"
      }
    ],
    "text": "CT dose and image quality category"
  },
  "subject": {
    "reference": "Patient/e1n5j3r6Goo1BGu7ckMTnSw3"
  },
  "focus": [
    {
      "reference": "DiagnosticReport/e9BvJll1zDW79iDWWfs2fTwXfc8vLgyjQb7F3I38cgZ43"
    }
  ],
  "encounter": {
    "reference": "Encounter/eknUIpa6enj2w4Kbl5CYa93rRgqcM7EeV5j.3hhPfshg3"
  },
  "effectiveDateTime": "2025-01-27T20:38:30Z",
  "valueCodeableConcept": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "LA31753-9",
        "display": "Abdomen and Pelvis Routine Dose"
      }
    ]
  },
  "component": [
    {
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "96912-1",
            "display": "Calculated CT global noise"
          }
        ]
      },
      "valueQuantity": {
        "value": 20.01,
        "unit": "[hnsf'U]",
        "system": "http://unitsofmeasure.org",
        "code": "[hnsf'U]"
      }
    },
    {
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "96913-9",
            "display": "Calculated CT size-adjusted dose"
          }
        ]
      },
      "valueQuantity": {
        "value": 237.54,
        "unit": "mGy.cm",
        "system": "http://unitsofmeasure.org",
        "code": "mGy.cm"
      }
    }
  ]
}
```


Sample response

HTTP 200